

Adapting Heterogeneity Aware Metrics to Tabular Health Data: A Numerical HTA Framework and Empirical Evaluation

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1 Introduction

Health datasets are often heterogeneous even within a single cohort. Patients can differ across demographic, behavioral, and clinical characteristics, and these differences may appear jointly rather than through any one variable alone. Measuring heterogeneity matters because it is closely related to challenges in machine learning: highly heterogeneous data may require more samples for stable training, while underrepresented subgroups can reduce generalization and predictive reliability. In this sense, heterogeneity is connected not only to class imbalance, but also to broader mismatch in feature distributions across subgroups.

However, heterogeneity in tabular health data is often summarized using single variable measures or global diversity indices. These summaries are useful, but they may miss local multivariate structure and do not directly capture how combinations of patient traits vary across the covariate space. This is a limitation because model performance often depends on joint patterns rather than isolated variables.

The HeTerogeneity Average, or HTA, was originally developed to measure multivariate heterogeneity across spatial regions and then aggregate local heterogeneity into a global score. Its appeal is that it is multivariate, bounded between 0 and 1, and based on local composition rather than only global summaries. In this project, I adapt HTA to tabular health data by defining regions in covariate space and measuring heterogeneity through joint trait combinations. I then compare this numerical HTA framework with benchmark measures such as alpha Simpson diversity, and examine whether it provides a useful summary of data complexity for machine learning settings.

2 Methodological background

2.1 Original HTI and HTA

The original HTA framework is built on the HeTerogeneity Index, or HTI, which measures heterogeneity within a single region using a normalized entropy formulation. Suppose a region contains observations belonging to C possible non empty trait combinations, and let p_c denote the proportion of observations in that region that belong to trait combination c . Then HTI is defined as

$$\text{HTI} = - \sum_{c=1}^C p_c \log_C(p_c).$$

Because the logarithm is taken with base C , HTI is bounded between 0 and 1. An HTI value close to 0 indicates that one trait combination dominates the region, corresponding to low local heterogeneity. An HTI value close to 1 indicates that the trait combinations are more evenly represented, corresponding to high local heterogeneity. In the original paper, HTI was introduced as a variation of Shannon entropy for measuring heterogeneity in trait combination data.

While HTI captures heterogeneity within a single region, it does not account for how heterogeneity varies across different regions of the sample. To address this, the HeTerogeneity Average, or HTA, is defined as a weighted average of local HTI values across a collection of regions. Regions with more observations receive greater weight, so the global score reflects both within region heterogeneity and the relative size of each region. As described in the original HTA paper, this allows HTA to summarize sample level heterogeneity while preserving local structure.

In this sense, HTI is a local measure, whereas HTA is a global aggregation of local heterogeneity. This distinction is important because two datasets may have similar overall trait proportions but differ substantially in how those trait combinations are distributed across regions. The HTA framework was designed specifically to capture this local to global structure.

2.2 Numerical adaptation idea

To adapt HTI and HTA to tabular health data, this project replaces spatial regions with regions defined in covariate space. In this report, Numerical HTA refers to the adaptation of HTA to tabular, non imaging health data through regions defined in numerical covariate space. This does not require all trait definitions to remain fully continuous. Instead, the framework allows trait definitions to vary across analyses while preserving the same region based logic. In the main analysis, Age and BMI are used to define these regions. For a chosen value of k , each variable is divided into k bins, producing k^2 Age by BMI regions. Local heterogeneity is then evaluated within each nonempty region.

The notion of a trait is also redefined for the tabular setting. Instead of using spatial trait maps, each individual is assigned to a joint trait category formed by combining selected variables of interest. In the Alzheimer dataset, the main trait definition is based on the joint combination of Gender, Ethnicity, EducationLevel, and Smoking. Within each Age by BMI region, the empirical distribution of these joint trait categories is used to compute local HTI.

The local HTI values are then aggregated using region size weights to obtain a global Numerical HTA score. This preserves the main logic of the original HTA framework while redefining space in terms of covariate neighborhoods rather than physical locations. The resulting score provides a bounded summary of multivariate heterogeneity that can be compared across grid resolutions and datasets.

2.3 Benchmark measures and comparative framework

Because Numerical HTA is adapted from a method originally developed for spatial data, it is useful to compare it with a more standard diversity measure in the tabular setting. In this project, the main benchmark is alpha Simpson diversity, computed under the same Age by BMI regional structure as Numerical HTA. This comparison helps assess whether Numerical HTA behaves mainly like a conventional within region diversity summary or whether it captures something more specific

to joint trait composition.

The comparison with alpha Simpson diversity is used to evaluate how closely HTA tracks within region mixture diversity across datasets and trait definitions. Since both measures are based on the distribution of trait categories within each region, their similarity or difference helps clarify what Numerical HTA is capturing in practice. In the final report, this benchmark serves as the main reference point for interpreting the empirical behavior of Numerical HTA.

3 Data

3.1 Alzheimer’s disease cohort

The main empirical analysis uses an Alzheimer’s disease tabular cohort as the primary real data example. According to the original project proposal, this cohort contains 2,149 patients and includes both numerical variables, such as age and BMI, and categorical variables, such as sex, ethnicity, education, and smoking. In the final Numerical HTA pipeline, this dataset is used to define Age by BMI regions and multivariate trait combinations from selected categorical variables.

This dataset was chosen because it provides a natural setting for adapting HTA from spatial data to tabular health data, with numerical variables defining covariate space and categorical variables defining trait composition within each region.

3.2 Comparison datasets

In addition to the Alzheimer’s cohort, the project uses comparison datasets to evaluate whether Numerical HTA behaves as expected under different levels of heterogeneity. In the final analysis, a pseudo homogeneous dataset is used as a contrast to the more heterogeneous reference data.

3.3 Numerically dominated dataset for the PCA tertile extension

For the PCA based extension, the project also considers a more numerically dominated dataset, referred to in the project summary as the MultiNumeric Cohort. This dataset is used because PCA is more suitable when the feature set is dominated by numerical variables, and it contains 18,418 patients. It supports the extension analysis in which principal components are used to construct alternative trait definitions, allowing assessment of how sensitive Numerical HTA is to representation choices.

4 Methods

4.1 Data preprocessing

For the main Alzheimer analysis, the data were prepared by ensuring that the variables used in the Numerical HTA pipeline were in the correct format before heterogeneity was computed. Age and BMI were retained as continuous variables for region construction, while the trait variables Gender, Ethnicity, EducationLevel, and Smoking were converted to factors so that each individual could be assigned to a joint categorical trait profile. Regions were then defined by partitioning Age and BMI into quantile based bins. For a chosen value of k , each variable was divided into k groups, producing an Age by BMI grid with k^2 cells. Empty cells could occur at finer resolutions, so HTA was computed from the nonempty cells.

For the PCA tertile extension, preprocessing focused on the numerically dominated dataset. Rows with missing values were removed, and unnecessary columns were dropped where appropriate. Age and BMI were excluded from PCA so that the variables used for region definition were not reused in trait construction. The remaining numeric variables were then standardized before principal component analysis. The resulting principal component scores were used to construct alternative trait definitions, while the Age by BMI grouping was retained as the region definition. This preprocessing design allowed the project to compare the original Age by BMI based Numerical HTA pipeline with a PCA based trait extension under a consistent framework.

4.2 Numerical HTA on the Alzheimer dataset

The main Numerical HTA analysis was conducted on the Alzheimer cohort using Age by BMI regions and multivariate categorical trait combinations. Within each nonempty region, local heterogeneity was computed from the empirical distribution of joint trait categories.

Trait categories were defined by the combination of four categorical variables: Gender, Ethnicity, EducationLevel, and Smoking. Each individual was therefore assigned to one multivariate trait profile, and the local HTI for a region was computed from the distribution of these profiles within that region. Global Numerical HTA was then obtained by aggregating the local HTI values across regions using region size weights.

HTA was evaluated across multiple grid resolutions to examine how the measured heterogeneity changed under finer or coarser partitions of the covariate space. The Alzheimer dataset served as the main empirical setting because it naturally supports this framework: Age and BMI define the regional structure, while the selected categorical variables define trait composition within each region.

In the final project, however, this component remained exploratory and was not developed into a fully reported inferential analysis. The main conclusions of the report therefore rely on descriptive patterns across datasets, grid resolutions, and benchmark comparisons, rather than on permutation based significance claims.

4.3 Visualizing within cell trait composition

To support interpretation of Numerical HTA, the project also examined within region trait composition using stacked proportion plots. These visualizations show how the joint trait categories are distributed across Age by BMI regions and help clarify what HTA is measuring locally.

This step was used mainly as an interpretive aid rather than as a separate inferential analysis. In the report, these plots help connect the numerical HTA values to the underlying composition patterns within regions.

4.4 Benchmark comparison: alpha Simpson

A second benchmark comparison used alpha Simpson diversity, computed within regions and then summarized across the same Age by BMI partition. This provided a standard diversity based measure for comparison with Numerical HTA under the same regional structure.

The purpose of this benchmark was to assess whether Numerical HTA behaved similarly to an established within region mixture measure. In the final report, alpha Simpson is used mainly as a comparative reference rather than as a separate main analysis.

4.5 PCA based HTA extension

As an exploratory extension, the project also examined a PCA based version of Numerical HTA on a numerically dominated dataset. In this extension, Age and BMI were retained as the region defining variables, while principal components derived from the remaining numeric variables were used to construct alternative trait definitions. Age and BMI were excluded from the PCA input so that the same variables were not used both for region definition and for downstream trait construction.

The final PCA based analysis focused on a simplified representation using PC1 only. PC1 was discretized into tertiles, labelled low, mid, and high, and these three categories were treated as the trait definition within each Age by BMI region. Local HTI and global HTA were then computed from the within region distribution of these PC1 tertile categories.

This PCA tertile analysis was treated as an exploratory extension rather than part of the primary Alzheimer pipeline. Its purpose was to assess how Numerical HTA behaves when the trait definition is based on a latent numerical representation rather than on observed categorical variables.

5 Results

5.1 Numerical HTA with categorical traits

The main Numerical HTA analysis was carried out on the Alzheimer dataset using Age and BMI to define regions and using the joint combination of Gender, Ethnicity, EducationLevel, and Smoking to define categorical traits. Under this setup, each individual was assigned to exactly one multivariate trait category, and local heterogeneity was computed from the distribution of these trait categories within each Age by BMI cell. This formulation stays closest to the original HTA idea because it preserves an explicit trait composition within each region.

As shown in Figure A1 in Appendix, across grid resolutions, Numerical HTA presented a clear resolution effect. As the number of Age and BMI bins increased, the global HTA increased monotonically, indicating that finer partitions of the covariate space tended to produce higher measured heterogeneity under this categorical trait construction. This pattern suggests that the measure is sensitive to how finely the data are divided into local regions, and that more granular region definitions can amplify the apparent heterogeneity of the cohort.

In Figure A2, the within region composition plots help explain this behavior. In the representative 2 by 2 Age by BMI grid, each cell contained a mixture of joint trait categories rather than a single dominant profile, and the stacked composition plots provided a direct visual link between local trait balance and the resulting HTA values. Taken together, these results show that the categorical trait version of Numerical HTA yields an interpretable summary of multivariate heterogeneity in tabular health data and provides the clearest baseline for the later comparisons in this report.

5.2 Comparison with a more homogeneous dataset

To examine whether Numerical HTA could distinguish different levels of heterogeneity under the same regional framework, the main categorical trait analysis was compared with a pseudo homogeneous dataset constructed under the same Age by BMI grid design. In this comparison, the heterogeneous setting used joint categorical traits based on the original Alzheimer analysis, whereas the homogeneous setting used a simpler trait construction that produced a more concentrated within cell composition.

Under this comparison, the categorical trait version of Numerical HTA behaved in the expected direction. In Figure A3, the more homogeneous dataset showed lower heterogeneity, and the within cell composition plots were consistent with this result: each Age by BMI region was dominated by a smaller number of trait categories, rather than showing the broader mixture seen in the more heterogeneous setting. This suggests that, when traits are defined through observed categorical combinations, Numerical HTA can reflect meaningful differences in local trait concentration across datasets.

This comparison is informative, but it does not isolate heterogeneity alone, since the trait construction also differs between the two settings. Among the retained analyses, the categorical trait formulation still stayed closest to the original HTA logic and gave the most interpretable heterogeneous versus homogeneous ordering. For that reason, it serves as the main reference point for the benchmark comparison in the next subsection and for the PCA tertile extension discussed later.

5.3 Comparison with alpha Simpson diversity

Figure A4 compares Numerical HTA with alpha Simpson diversity across Age by BMI grid resolutions for both the heterogeneous and homogeneous datasets. In the heterogeneous dataset, the two measures followed closely related patterns across all three grid settings. Alpha Simpson remained slightly higher than HTA at each resolution, but both measures stayed consistently high.

A similar correspondence appeared in the homogeneous dataset. Both measures were much lower than in the heterogeneous case, and both peaked at the 3 by 3 grid before decreasing again at the 4 by 4 grid. HTA was slightly higher than alpha Simpson at each resolution in this setting, but the overall shape of the curves remained very similar.

Taken together, Figure A4 shows that alpha Simpson diversity and Numerical HTA followed closely matched empirical patterns across the retained settings, with the main difference being a modest shift in level rather than a change in overall shape.

5.4 PCA tertile extension

As an exploratory extension, Numerical HTA was also evaluated on a numerically dominated dataset using a PCA based trait construction. In this analysis, Age and BMI were retained as the region defining variables, while PC1 from the remaining standardized numeric variables was divided into three tertile based categories labelled low, mid, and high. These PC1 tertile labels were then treated as the trait definition within each Age by BMI region.

Under this setup, HTA remained very high in both datasets, and the expected heterogeneous versus homogeneous ordering did not hold. In particular, the homogeneous dataset could produce an HTA

value that was as high as or higher than that of the heterogeneous dataset. This means that the PC1 tertile extension did not reliably recover the intended heterogeneous versus homogeneous distinction.

The observed pattern suggests that this branch was highly sensitive to the PCA based trait definition. Rather than serving as a stable main analysis, it is better interpreted as an exploratory sensitivity check. A summary of the retained analyses and their interpretations is provided in Table A1 in the Appendix.

6 Discussion

6.1 Main interpretation of the categorical trait analysis

The categorical trait analysis gave the clearest and most interpretable adaptation of HTA to tabular health data in this project. When Age and BMI were used to define regions and the joint combination of Gender, Ethnicity, EducationLevel, and Smoking was used to define traits, Numerical HTA stayed closest to the original local to global logic of HTA. In this setting, each region had an explicit trait composition, and the global score summarized how these compositions varied across the Age by BMI grid.

This formulation also produced the most sensible heterogeneous versus homogeneous comparison. The categorical trait version showed lower heterogeneity in the more homogeneous setting, which was the expected direction and made this branch the most interpretable comparative result in the report. This suggests that, when traits are defined through observed categorical combinations rather than latent summaries, Numerical HTA can distinguish meaningful differences in within region trait concentration.

6.2 Comparison with alpha Simpson diversity

The comparison with alpha Simpson diversity helps clarify what Numerical HTA was capturing in the retained categorical trait analysis. Because the two measures showed closely matched patterns across grid resolutions in both datasets, the main empirical signal captured by Numerical HTA in this setting appears to be closely related to within region trait diversity.

This does not make HTA redundant. HTA still preserves the original local to global framework and provides a bounded score on a 0 to 1 scale. However, in the present tabular adaptation, its empirical behavior was not sharply distinct from that of a standard diversity index under the retained categorical trait definition.

6.3 Sensitivity to PCA based trait representation

The PCA tertile extension showed that Numerical HTA was highly sensitive to how traits were defined. When traits were constructed from PC1 tertiles rather than from observed categorical combinations, HTA remained very high and could even be higher for the more homogeneous dataset. This happened because the trait definition itself introduced an artificial form of heterogeneity. By dividing PC1 into low, mid, and high tertiles, the analysis created three categories that were relatively balanced by construction, which made within region trait mixtures appear highly

heterogeneous. Although balanced tertile mixing is a plausible explanation, it does not fully resolve the empirical behavior observed in this branch.

At the same time, the Age and BMI grid did not appear to separate these PCA based traits in a way that produced strong local concentration. Instead, the tertile categories remained fairly evenly mixed within many regions, especially in the homogeneous dataset. As a result, local HTI values stayed high across the grid, which in turn kept global HTA high. In this setting, the measure reflected balanced mixing of latent categories more than the intended heterogeneous versus homogeneous distinction.

This makes the PCA tertile branch useful as a sensitivity analysis, but less interpretable than the categorical trait analysis. It also points to an unresolved issue in the current framework, namely that discretizing latent components can itself induce high apparent heterogeneity. This motivates the future directions discussed later, where alternative trait constructions may help make the extension more interpretable.

6.4 Limitations and future work

This project has several limitations. First, the results depended strongly on how regions and traits were defined. In the retained analysis, Age and BMI were used to define the regional grid, while trait definitions were based either on observed categorical combinations or on a PCA tertile construction. As the PCA extension showed, changing the trait representation could materially change the behavior of Numerical HTA and even weaken the intended heterogeneous versus homogeneous ordering.

Second, the main analysis remained descriptive rather than fully inferential. The report focused on empirical patterns across datasets, grid resolutions, and benchmark comparisons, rather than on formal significance testing. This makes the current results useful for interpretation and comparison, but not yet a complete inferential framework.

A related limitation is that the PCA tertile extension may have introduced apparent heterogeneity through the discretization step itself. Dividing a latent component into tertiles can create balanced categories by construction, which may push HTA upward even when the underlying dataset is relatively homogeneous. Future work could address this by replacing entropy based comparison of within region trait balance with a divergence based approach that compares each cell to the global trait distribution. One possible direction, suggested during the project, is a Jensen Shannon based HTA in which each Age by BMI cell is compared against the global PC1 tertile distribution rather than being scored only by its internal balance. This could help distinguish genuinely divergent subgroup structure from heterogeneity that is induced by the tertile construction itself.

More broadly, future work could examine alternative trait constructions, different region definitions, and extensions to continuous representations that avoid hard discretization. These directions would help clarify when Numerical HTA provides information beyond a standard diversity index and when it is most useful as a heterogeneity summary for machine learning applications.

7 Conclusion

This project adapted the HTA framework from spatial data to tabular health data by defining regions in covariate space and measuring local heterogeneity through joint trait composition. In the main analysis, Age and BMI were used to define regions, while the joint combination of Gender, Ethnicity, EducationLevel, and Smoking defined categorical traits. Under this formulation, Numerical HTA gave the clearest and most interpretable results, and the comparison with a more homogeneous dataset followed the expected direction, with lower heterogeneity in the more homogeneous setting.

The benchmark comparison showed that alpha Simpson diversity closely tracked Numerical HTA across the retained settings. This suggests that, in the categorical trait analysis, the main empirical signal captured by Numerical HTA was closely related to within region trait diversity. At the same time, HTA remained useful because it preserved the original local to global framework and provided a bounded score on a 0 to 1 scale that was easy to interpret across settings.

The PCA tertile extension showed that Numerical HTA was sensitive to trait definition. When traits were constructed from a latent PCA representation, the resulting HTA values could remain high even for the more homogeneous dataset, making that version harder to interpret. Overall, the project shows that HTA can be meaningfully adapted to tabular health data, but also that its interpretation depends strongly on how regions and traits are defined. In this report, the categorical trait formulation provided the most coherent and useful version of Numerical HTA.

8 Appendix

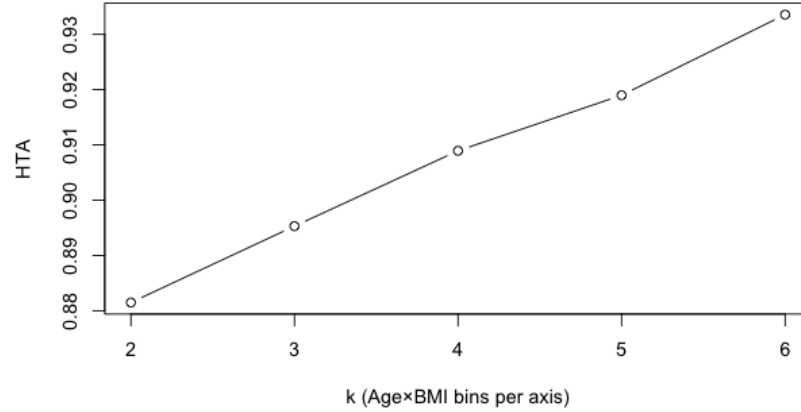


Figure A1: Global HTA as a function of the number of Age×BMI bins per axis, where k defines a $k \times k$ regional grid. Using the Alzheimer’s dataset, regions were constructed from Age and BMI, and trait categories were defined by the joint combinations of Gender, Ethnicity, EducationLevel, and Smoking. The figure shows a monotone increase in HTA as k increases from 2 to 6, indicating that finer regional discretization yields higher measured heterogeneity under this Numerical HTA framework. This pattern is consistent with a resolution effect, where more granular subgroup definitions reveal greater within dataset variation.

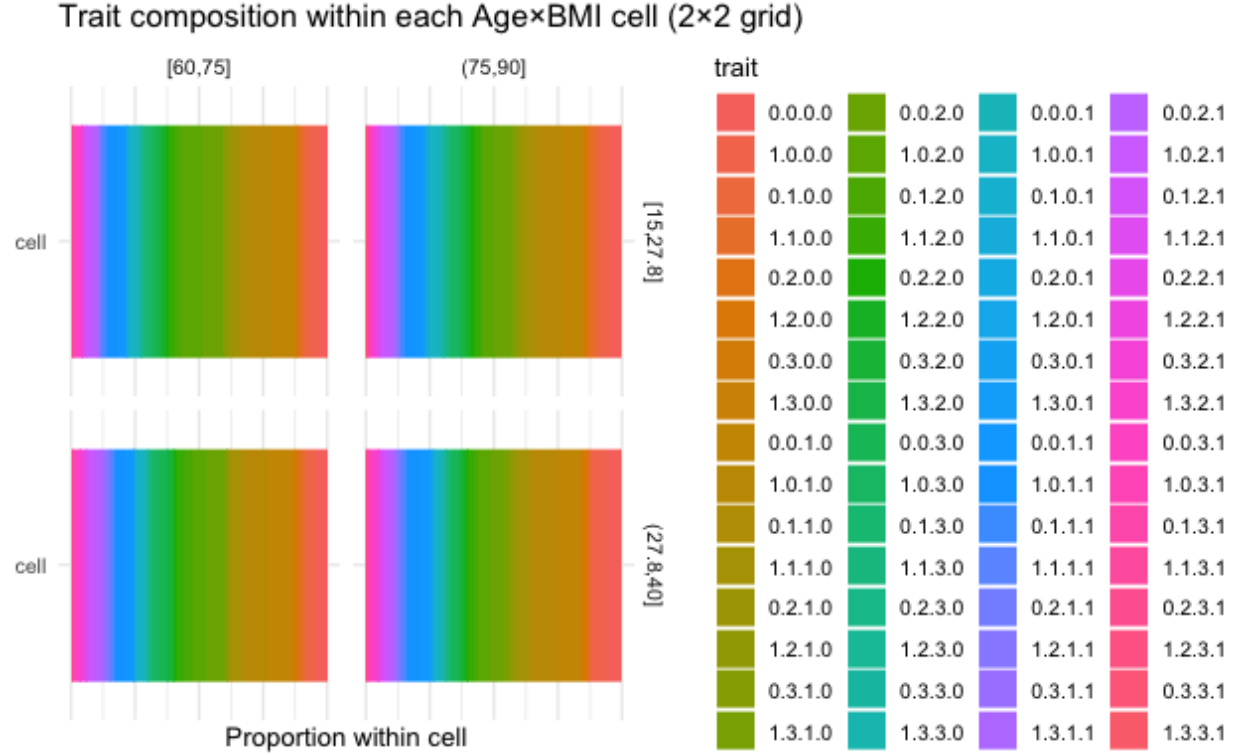


Figure A2: Trait composition within each Age×BMI region under the 2×2 grid definition in the Alzheimer’s dataset. Regions are formed by crossing two Age bins with two BMI bins, and each bar shows the within cell proportion of joint trait categories defined by Gender, Ethnicity, EducationLevel, and Smoking. The broadly similar but non identical compositions across the four cells indicate that the dataset contains substantial multivariate trait diversity within each region, while also preserving between region differences in joint trait distribution. This cell level composition is the basis for the local HTI values that are then aggregated into the global HTA.

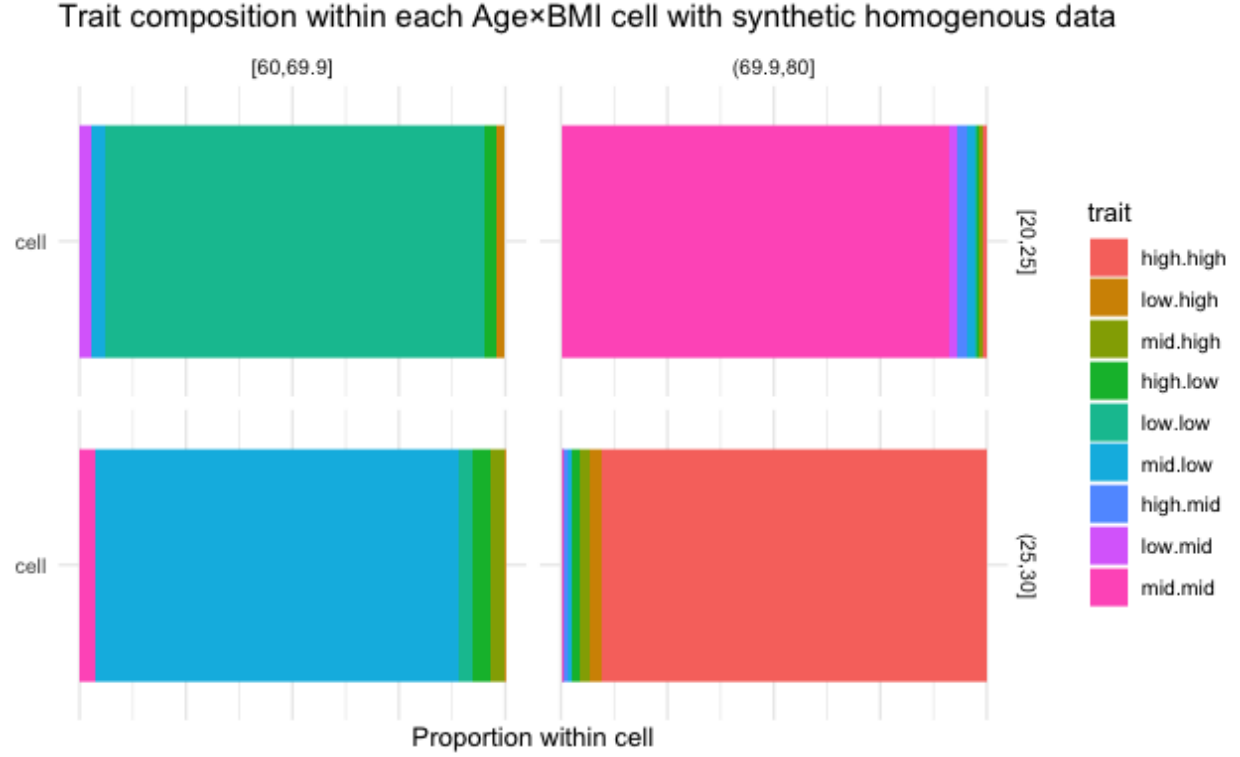


Figure A3: Trait composition within each Age×BMI region for the pseudo homogeneous dataset under the 2×2 grid definition. Regions are formed by crossing two Age bins with two BMI bins, and trait categories are defined by the joint combinations of SBP group and LDL group. Unlike the Alzheimer’s dataset, each cell here is dominated by a single trait category, with only small contributions from the remaining categories. This strong within cell concentration is consistent with the dataset being constructed to represent a more homogeneous joint trait structure across Age×BMI regions.

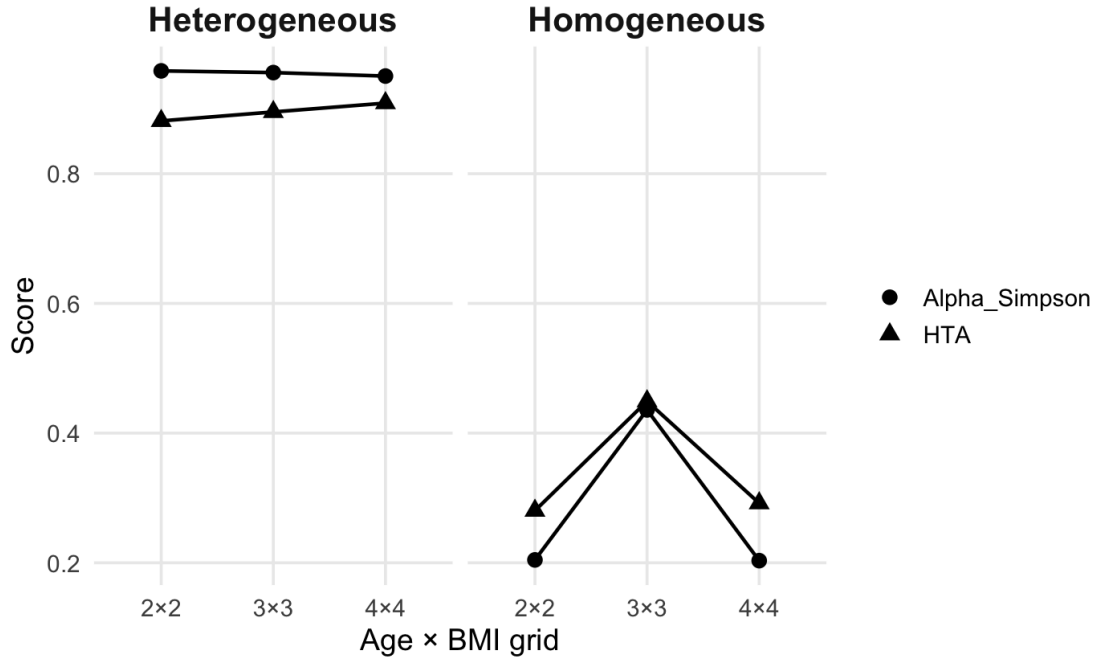


Figure A4: Comparison of HTA and the alpha Simpson index across Age×BMI grid resolutions for the heterogeneous and homogeneous pseudo datasets. Both metrics show very similar patterns within each dataset. In the heterogeneous dataset, HTA and alpha Simpson remain high and relatively stable across the 2×2 , 3×3 , and 4×4 grids. In the homogeneous dataset, both metrics are substantially lower overall, with a local peak at the 3×3 resolution. This figure suggests that, in this setting, HTA tracks cell level trait mixture diversity in a manner similar to the alpha Simpson index.

Table A1: Summary of HTA analyses retained for the final report.

Approach	Dataset	Region definition	Number of traits	Main finding	Interpretation
Categorical traits	Heterogeneous dataset	Age $\times$ BMI grid	64	HTA remained high under categorical trait composition.	Most faithful to the original HTA setting.
Categorical traits	Homogeneous dataset	Age $\times$ BMI grid	9	HTA was lower and within cell compositions were more concentrated.	Sensitive to meaningful trait concentration differences.
PCA tertile traits	Numerical dataset	Age $\times$ BMI grid	3	HTA remained high and could be higher for the homogeneous dataset.	Captures latent balance more than raw heterogeneity.
Simpson alpha	Same setup as HTA comparison	Age $\times$ BMI grid	Same as corresponding trait definition	Simpson alpha closely tracked HTA across settings.	HTA largely reflected standard within cell diversity.